choreographerManagement

Service Description

Service ID: *"choreographerManagement"*

Abstract
This document describes the `choreographerManagement` service, version 1.0.0, produced by the Choreographer support system of Arrowhead 5.2.2.

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1 Overview

This document describes the `choreographerManagement` service, which provides operator management of plans, sessions, session steps, the worklog and executors. It is produced by the Choreographer system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

1.1 Significant Prior Art

Successor of the Arrowhead 4.6 Choreographer management endpoints.

1.2 How This Service Is Meant to Be Used

Operators create, check and remove plans, start and abort sessions, follow sessions, steps and the worklog, and manage executors.

1.3 Important Delimitations and Dependencies

Access follows the management policy (standalone deployment: `sysop-only`).

2 Service Interface

This section describes the interfaces to the `choreographerManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

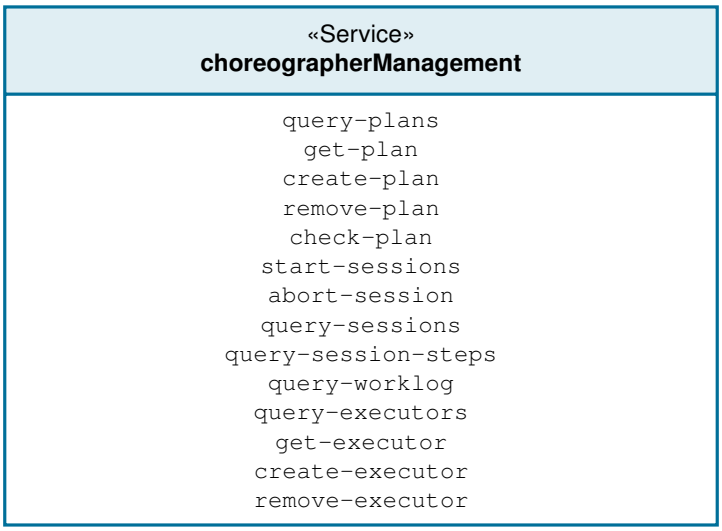


Figure 1: SysML-style block of the `choreographerManagement` service and its operations.

The following interface operations are available.

2.1 operation query-plans () : PlanListResponse

Returns a page of plans.

2.2 operation get-plan (Identifier) : PlanResponse

Returns a plan.

2.3 operation create-plan (PlanRequest) : PlanResponse

Validates and stores a plan (cycles, reachability, transitive reduction).

2.4 operation remove-plan (Identifier)

Removes a plan.

2.5 operation check-plan (Identifier) : CheckPlanResponse

Checks whether providers and executors exist for every step.

2.6 operation start-sessions (RunPlanListRequest) : RunPlanListResponse

Starts sessions of plans by identifier.

2.7 operation abort-session (Identifier)

Aborts a session.

2.8 operation query-sessions () : SessionListResponse

Returns a page of sessions.

2.9 operation query-session-steps () : SessionStepListResponse

Returns a page of session steps.

2.10 operation query-worklog () : WorklogListResponse

Returns a page of worklog entries.

2.11 operation query-executors () : ExecutorListResponse

Returns a page of executors.

2.12 operation get-executor (Identifier) : ExecutorResponse

Returns an executor.

2.13 operation create-executor (ExecutorCreateRequest) : ExecutorResponse

Registers an executor on behalf of a system.

2.14 operation remove-executor (Identifier)

Removes an executor.

3 Information Model

Here, all data objects that can be part of the `choreographerManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.25, which are used to represent things like hashes and identifiers.

3.1 struct ActionRequest

An action of a plan.

Field	Type	Description
name	Name	Name of the action within the plan.
nextActionName	String	Action executed after this one.
firstStepNames	List<String>	Steps that start the action.
steps	List<StepRequest>	The steps of the action.

3.2 struct ActionResponse

An action of a plan.

Field	Type	Description
id	Number	Identifier of the action.
name	Name	Name of the action within the plan.
firstAction	Boolean	Whether this is the plan's first action.
nextActionName	String	Action executed after this one.
firstStepNames	List<String>	Steps that start the action.
steps	List<StepResponse>	The steps of the action.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.3 struct CheckPlanResponse

Result of checking a plan against the current cloud.

Field	Type	Description
planId	Number	The checked plan.
errorMessages	List<String>	Problems found in the plan.
needInterCloud	Boolean	Whether the plan needs providers in other clouds.

3.4 struct ExecutorCreateRequest

An executor to create on behalf of a system.

Field	Type	Description
systemName	Name	Executor system.
address	String	One network address (host name, IPv4, IPv6 or MAC address).
port	Number	Port of the executor interface.
basePath	String	Base path of the executor interface.
serviceDefinitionName	Name	Service definition the executor can execute steps for.
minVersion	Number	Lowest accepted major version.
maxVersion	Number	Highest accepted major version.

3.5 struct ExecutorListResponse

List of executors.

Field	Type	Description
entries	List<ExecutorResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.6 struct ExecutorResponse

A registered executor.

Field	Type	Description
id	Number	Identifier of the executor.
name	Name	Executor system.
address	String	Address of the executor interface.
port	Number	Port of the executor interface.
basePath	String	Base path of the executor interface.
serviceDefinitions	List<ExecutorServiceDefinitionResponse>	Service definitions the executor serves.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.7 struct ExecutorServiceDefinitionResponse

A service definition an executor serves.

Field	Type	Description
serviceDefinitionName	Name	Service definition name, camelCase (e.g. `temperatureReading`).
minVersion	Number	Lowest accepted major version.
maxVersion	Number	Highest accepted major version.

3.8 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

3.9 struct OrchestrationServiceRequirement

Requirements on the requested service.

Field	Type	Description
serviceDefinition	Name	Service definition name.
operations	List<String>	Service operations the consumer wants to use.
versions	List<Version>	Accepted service versions.
alivesAt	DateTime	Only instances that are still registered (not expired) at this UTC time are matched.
metadataRequirements	List<MetadataRequirement>	Requirements on the service instance metadata; each map is one alternative.
interfaceTemplateNames	List<String>	Accepted interface templates, e.g. `generic_https`.
interfaceAddressTypes	List<String>	Address types the provider interface must offer.
interfaceProperty Requirements	List<MetadataRequirement>	Requirements on the interface properties; each map is one alternative.
securityPolicies	List<String>	Accepted interface security policies.
preferredProviders	List<String>	Provider systems to prefer.

3.10 struct PlanListResponse

List of plans.

Field	Type	Description
entries	List<PlanResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.11 struct PlanRequest

A plan to create.

Field	Type	Description
name	Name	Unique plan name.
firstActionName	String	Action the plan starts with.
actions	List<ActionRequest>	The actions of the plan.

3.12 struct PlanResponse

A stored plan.

Field	Type	Description
id	Number	Identifier of the plan.
name	Name	Unique plan name.
firstActionName	String	Action the plan starts with.
actions	List<ActionResponse>	The actions of the plan.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.13 struct RunPlanListRequest

Sessions to start by plan identifier.

Field	Type	Description
sessions	List<RunPlanRequest>	Sessions to start or return.

3.14 struct RunPlanListResponse

Started sessions.

Field	Type	Description
entries	List<RunPlanResponse>	Matching records of the current page.

3.15 struct RunPlanRequest

Start of a session of a plan.

Field	Type	Description
planId	Number	Identifier of the plan.
quantity	Number	Number of times the plan (or step) is executed.
allowInterCloud	Boolean	Allow inter-cloud orchestration for the steps.
chooseOptimalExecutor	Boolean	Rank the local executors and choose the best one.
notifyProtocol	String	Protocol of session notifications, e.g. `HTTPS`.
notifyAddress	String	Address of session notifications.
notifyPort	Number	Port of session notifications.
notifyPath	String	Path of session notifications.

3.16 struct RunPlanResponse

A started (or refused) session.

Field	Type	Description
planId	Number	Identifier of the plan.
sessionId	Number	Identifier of the session.
quantity	Number	Requested executions.
status	SessionStatus	Status of the session.
errorMessages	List<String>	Problems found in the plan.
needInterCloud	Boolean	Whether the plan needs providers in other clouds.

3.17 struct SessionListResponse

List of sessions.

Field	Type	Description
entries	List<SessionResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.18 struct SessionResponse

A session.

Field	Type	Description
id	Number	Identifier of the session.
planId	Number	Identifier of the plan.
planName	String	Name of the plan.
status	SessionStatus	Session status.
quantityDone	Number	Executions finished so far.
quantityGoal	Number	Executions requested.
executionNumber	Number	Number of the current execution.
owner	Name	System that started the session.
notifyUri	String	Where session notifications are sent.
startedAt	DateTime	UTC time when the job started.
updatedAt	DateTime	UTC time of the last change.

3.19 struct SessionStepListResponse

List of session steps.

Field	Type	Description
entries	List<SessionStepResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.20 struct SessionStepResponse

A step execution of a session.

Field	Type	Description
id	Number	Identifier of the session step.
sessionId	Number	Identifier of the session.
stepId	Number	Identifier of the plan step.
planName	String	Name of the plan.
actionName	String	Name of the action.
stepName	String	Name of the step.
executorName	String	Executor system running the step.
status	SessionStepStatus	Status of the session step.
executionNumber	Number	Number of the current execution.
message	String	Text of the log entry or status message of the job.
startedAt	DateTime	UTC time when the job started.
updatedAt	DateTime	UTC time of the last change.

3.21 struct StepRequest

A step of an action.

Field	Type	Description
name	Name	Name of the step within the action.
quantity	Number	Number of times the plan (or step) is executed.
serviceRequirement	OrchestrationServiceRequirement	Orchestration requirement the executor's provider must meet.
localCloudOnly	Boolean	Use only providers of the local cloud.
staticParameters	Map<String, String>	Parameters passed to the executor unchanged.
nextStepNames	List<String>	Steps that start when this one has finished.

3.22 struct StepResponse

A step of an action.

Field	Type	Description
id	Number	Identifier of the step.
name	Name	Name of the step within the action.
serviceDefinition	Name	Service definition of the step's requirement.
minVersion	Number	Lowest major version derived from `versions`.
maxVersion	Number	Highest major version derived from `versions`.
quantity	Number	Number of times the plan (or step) is executed.
serviceRequirement	OrchestrationServiceRequirement	Orchestration requirement the executor's provider must meet.
localCloudOnly	Boolean	Use only providers of the local cloud.
staticParameters	Map<String, String>	Parameters passed to the executor unchanged.
nextStepNames	List<String>	Steps that start when this one has finished.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.23 struct WorklogListResponse

List of worklog entries.

Field	Type	Description
entries	List<WorklogResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.24 struct WorklogResponse

A worklog entry.

Field	Type	Description
id	Number	Identifier of the worklog entry.
entryDate	DateTime	UTC time of the log entry.
planName	String	Name of the plan.
actionName	String	Name of the action.
stepName	String	Name of the step.
sessionId	Number	Identifier of the session.
executionNumber	Number	Number of the current execution.
message	String	Worklog text.
exception	String	Stack trace or exception text of the log entry, if any.

3.25 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
Boolean	One out of `true` or `false`.
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
SessionStatus	One of `INITIATED`, `RUNNING`, `ABORTED`, `DONE`.
SessionStepStatus	One of `WAITING`, `RUNNING`, `ABORTED`, `DONE`.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

4 References

[1] J. Delsing, “Choreographer – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-support-java-spring/Choreography/documentation/v5.2.2/Choreographer-v5.2.2-SysD.pdf.

[2] ——, “choreographerManagement HTTP/TLS/JSON – Interface Design Description (Choreographer),” Eclipse Arrowhead, IDD 5.2.2, 2026, choreographer-v5.2.2-choreographerManagement-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	